CONSULTING CASE · STORE PERFORMANCE EVALUATION

Is Store 10 actually a good store?

A model-based evaluation of Store 10's weekly sales performance — correcting three common misreadings of the data and isolating the signal that actually matters for management.

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TECHNICAL APPENDIX

Bobs_Report.ipynb (submitted separately)

DATE

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01 Executive Summary

> *Store 10 is a **healthy, above-expected performer**, ranking in the top decile of the 45-store network on a fair, size- and type-adjusted basis. The apparent year-over-year decline suggested by a naïve reading of the sales data **is not real** — it is a byproduct of comparing unequal data windows. The operational insight worth acting on lives one level deeper: inside the store, the department mix is quietly shifting.*

FOUR FINDINGS, AT A GLANCE

FINDING 01

Store 10 is about 2% above where a peer-equivalent store would land

A model trained on all 45 stores predicts what a store with Store 10's size, type, location conditions and season *should* sell each week. Store 10 consistently beats that prediction by **+2.1% on average**, placing it **7th of 45** on peer-adjusted performance.

FINDING 02

The "decline" in Store 10's annual numbers is a measurement artifact

2010 covers 48 weeks, 2011 covers 52, and 2012 covers only 43 — **missing November and December, the two biggest sales months**. On like-for-like windows (Feb–Oct every year), Store 10 changes by **-1.5%** then *+2.0%* — essentially flat.

FINDING 03

Store 10 held up better than its peers through the 2011 weakness

Indexing sales to 2010 = 100, Store 10 ended 2011 at **98.5** while its size-matched Type B peer cohort dropped to **95.1**. In 2012 Store 10 recovered fully to 100.6 while peers recovered to 96.7. The store is **more resilient than its benchmark**, not less.

FINDING 04

The real operational signal is in the department mix

A flat store-level number hides material internal movement. Five departments are gaining meaningfully (led by **Dept 38** at +13% on an $87K/week base) and five are losing ground (led by **Dept 5** losing roughly $8K/week off a $56K base). This is where the manager's attention should go.

WHAT THIS CHANGES

Bob's original submission would have handed the client the headline "*Store 10 is a high-performing but slowly declining store*". Both halves of that statement are wrong. The correct framing is "*Store 10 is a solidly above-expected, resilient store with a flat top line and active departmental reshuffling that deserves attention.*" The recommendations in Section 6 follow from that framing, not the original one.

02 The question, and how we answered it

THE QUESTION

The brief asked for an evaluation of **Store 10's weekly sales performance**. Bob's first attempt compared Store 10's average weekly sales (~$1.9M) to the all-stores average (~$1.0M) and concluded Store 10 was a "good performing store". The comparison is structurally unfair: Store 10 is a mid-to-large Type B store, and the all-stores average mixes in small Type C stores that sell a tenth as much. A store with above-average size will look like an over-performer under any raw comparison.

We therefore rephrased the question into something measurable and decision-relevant:

> **THE QUESTION WE ACTUALLY ANSWERED**
>
> Given Store 10's **characteristics** (size, store type, location-specific conditions) and the **context of each week** (holiday, season, macro environment), how do its actual sales compare to the sales we would *expect* from a peer-equivalent store under the same conditions — and is that gap changing over time?

HOW WE ANSWERED IT

We worked at the **store-week level** (45 stores × ~143 weeks ≈ 6,400 observations) using three datasets provided: store attributes, weekly sales, and weekly contextual features (temperature, fuel price, CPI, unemployment, holiday flag).

Against that data we built two predictive models — a linear regression (simple and interpretable) and a Random Forest (non-linear, handles seasonality and interactions). Both were trained on historical data and evaluated on a later holdout period. The Random Forest was markedly more accurate (mean absolute error ≈ $97K per store-week, vs. $225K for the linear model) and became the basis for the residual analysis that drives this report's findings.

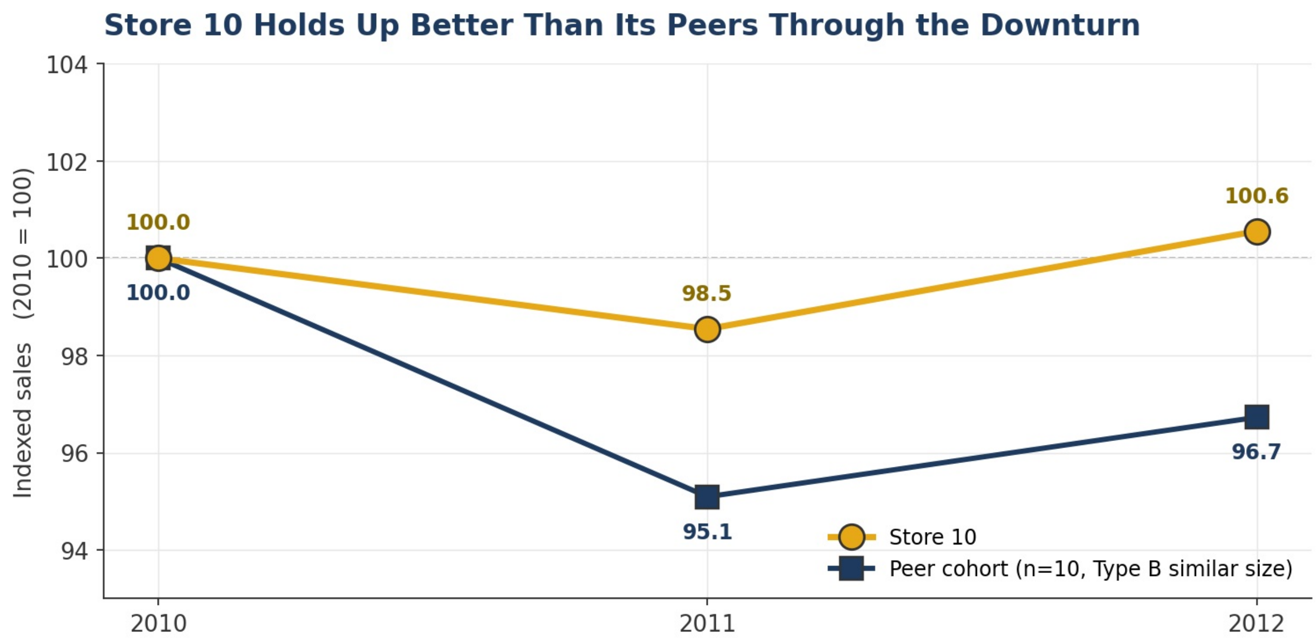
The model produces a peer-equivalent *expected* weekly sales figure for every store-week. The difference between actual and expected — the **residual** — is the clean, fair measure of over- or under-performance we use throughout.

Full modelling pipeline, metrics, train/test split, feature importances and diagnostics are documented in the companion technical notebook, Bobs_Report.ipynb.

03 Store 10 is a resilient, above-expected performer

HEADLINE **On a fair, size-matched basis, Store 10 tracks its peer cohort closely — and *outperformed* it through the 2011 weakness.**

A fair evaluation compares Store 10 to stores that are genuinely comparable: same store **Type**, similar **size**. Using Type B stores within ±25% of Store 10's 126,512 sq ft — a cohort of 10 peer stores — we indexed sales to 100 in 2010 and tracked each trajectory. Both series use the same Feb–Oct window every year so the comparison is clean.



Indexed weekly sales, 2010 = 100 (Feb–Oct window each year). Store 10 held up notably better than its peer cohort during the 2011 weakness (98.5 vs 95.1), and by 2012 had recovered fully to 100.6 while the cohort sat at 96.7. Peer cohort defined as Type B stores with size within ±25% of Store 10.

2011 INDEX — STORE 10

98.5

Lost just 1.5 points of index value during the downturn year.

2011 INDEX — PEER COHORT

95.1

The 10-store peer cohort lost more than twice that — 4.9 points.

PEER-ADJUSTED PERFORMANCE

+2.1%

Average weekly sales run 2.1% above the model's peer-equivalent prediction.

WHAT THIS MEANS

Being 2% above expectation might sound small, but in a 45-store network it is enough to place Store 10 **seventh** on peer-adjusted performance. More importantly, the 2011 peer comparison is a real stress test: every store in the cohort faced the same macro conditions (CPI, unemployment, fuel prices) and Store 10 responded better. Whatever the store is doing operationally, it is holding up through adverse conditions more effectively than comparable stores.

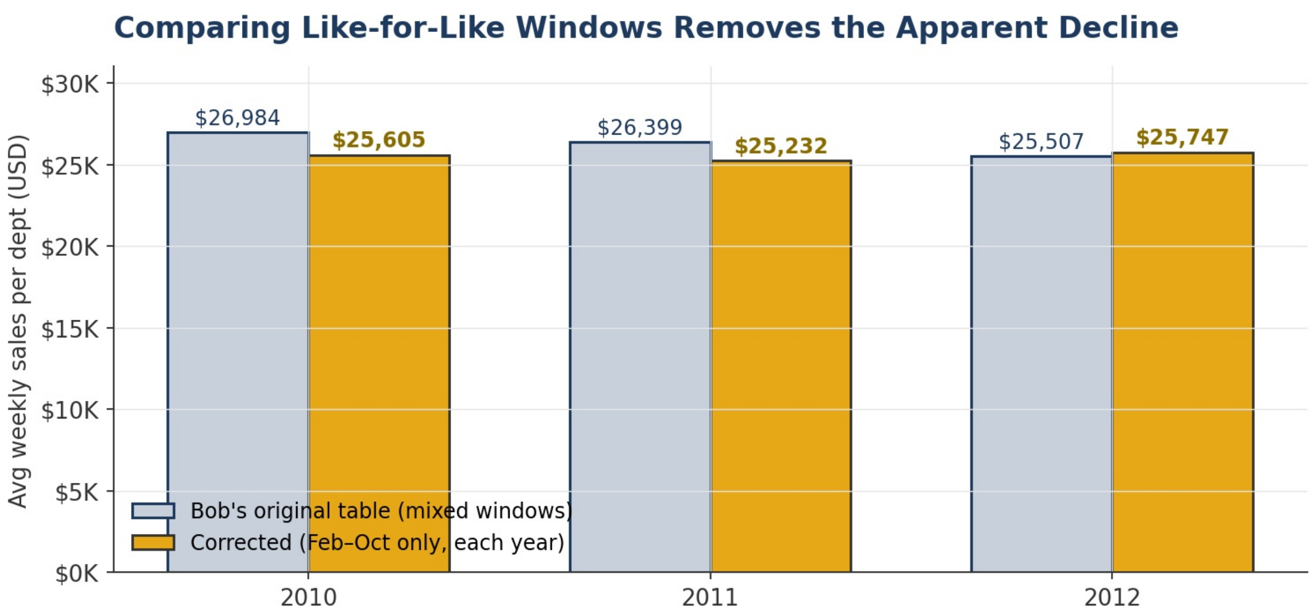
04 The apparent year-over-year decline is a measurement artifact

CORRECTION Bob's original Table 3 showed Store 10 falling ~5.5% from 2010 to 2012. On a like-for-like basis, the change is essentially zero.

A reviewer glancing at Bob's annual averages would see something alarming: $26,984 → $26,399 → $25,507. A gradually shrinking store is a very different story from "healthy performer", so before acting on that trend we have to confirm it is real. It turns out not to be — the three years cover very different windows:

YEAR	WEEKS OF DATA	PERIOD COVERED	MISSING FROM THE YEAR
2010	48	Feb 5 – Dec 31	First month of 2010
2011	52	Full year	None
2012	**43**	**Jan 1 – Oct 26**	**November & December — the two biggest sales months**

Any average that mixes these windows is comparing unequal baskets, and will make 2012 look smaller than it is. Restricting every year to the common Feb–Oct window removes the bias:



Avg weekly sales per department — Store 10, annual average. Bob's original table (light bars) mixes unequal windows and under-states 2012. The corrected series (gold bars) uses the same Feb–Oct window every year. On that basis the "decline" disappears: −1.5% from 2010 to 2011, and +2.0% from 2011 to 2012. Net over the period: +0.6%.

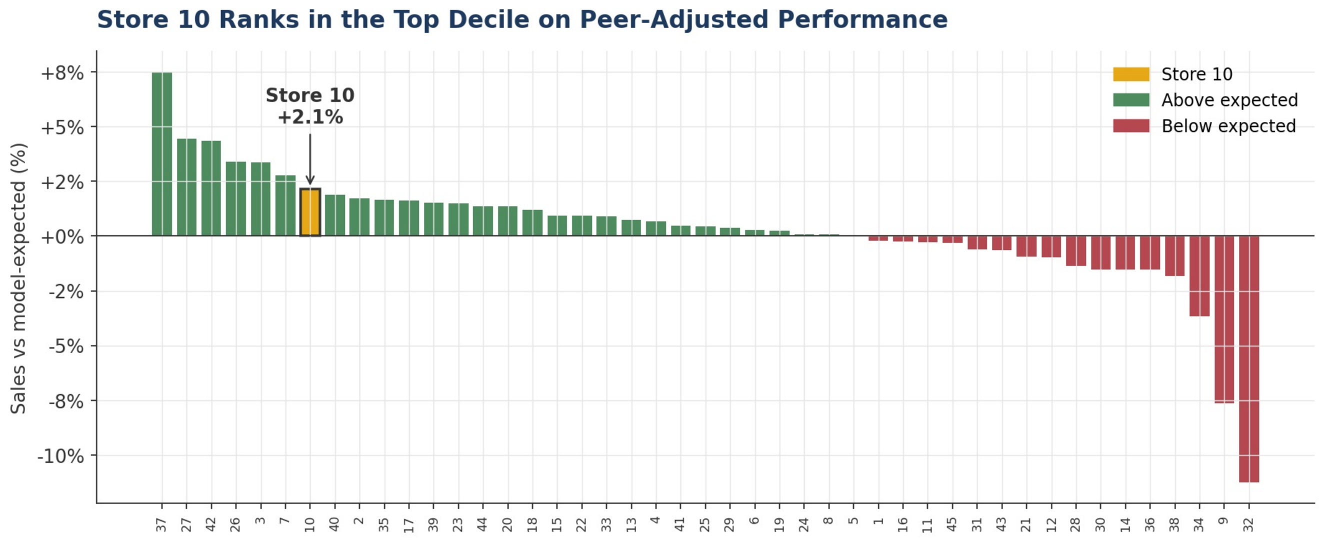
WHY THIS MATTERS BEYOND STORE 10

If Bob had presented his original numbers to a client, the first recommendation would likely have been "*investigate why Store 10 is shrinking.*" That investigation would have been chasing a measurement ghost — resources spent looking for a cause that was never there. The broader lesson is methodological: any year-over-year dashboard that compares *year-to-date* against *full-year* will produce the same phantom decline across many stores. A "comparable period mode" is a cheap, high-value addition to executive reporting.

05 Where Store 10 really ranks in the network

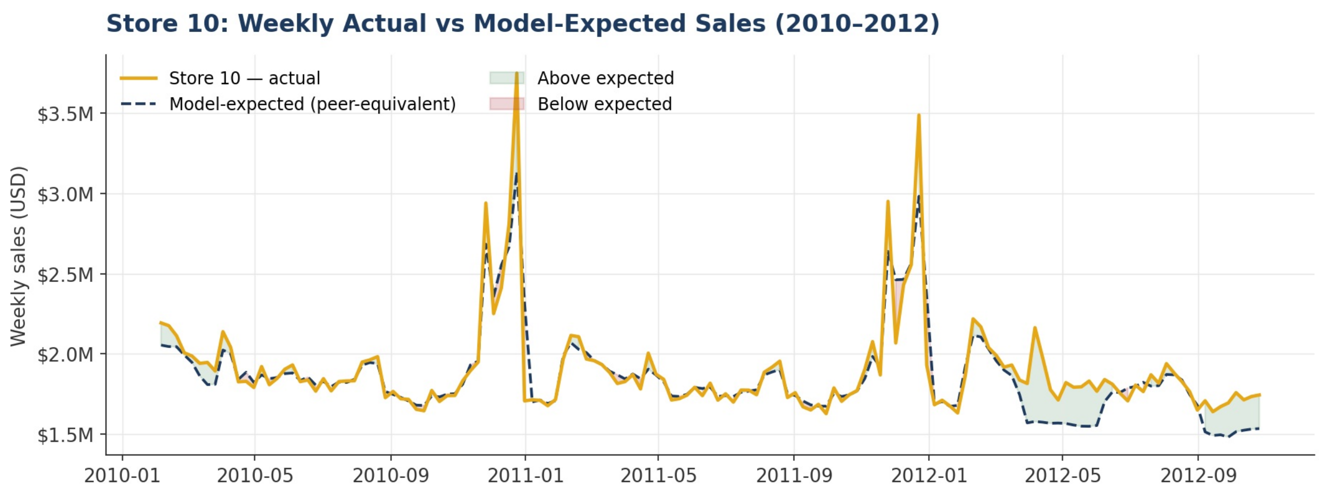
CONTEXT Solidly in the top decile of the network — strong, not an outlier.

Applying the model to every store-week and computing the residual gives us a clean performance-vs-expectation ranking for the whole network. Stores in green beat their peer-equivalent prediction; stores in red fall short; Store 10, highlighted in gold, lands **7th of 45**.



Average residual (actual minus model-expected), as % of predicted sales, by store. Positive bars are stores outperforming their peer-equivalent baseline; negative bars are underperforming. Store 10 is 7th of 45.

Week-by-week, Store 10's actual sales track the model's expected trajectory closely — the two lines overlap most of the time, which is another way of seeing that the store is behaving very much like a peer-equivalent store should. The gap, when it opens, opens *upward* (green shading below), especially around the Nov-Dec holiday peaks of 2010 and 2011.



Store 10 actual weekly sales (gold) vs model-expected (navy dashed). Holiday peaks in Nov-Dec 2010 and 2011 are the periods of greatest over-performance. The small period of underperformance in mid-2012 is one concrete thing to ask the store manager about.

06 Even with a flat top line, the department mix is shifting

OPERATIONAL Five departments are actively growing, five are actively shrinking. This is where the manager's attention belongs.

A stable store-level number can mask real internal movement. Decomposing Store 10's Feb–Oct average weekly sales by department — restricting to the 54 departments with at least $5K/week in average sales so small-base swings don't dominate — reveals the internal picture:

DEPARTMENTS GROWING 2010 → 2012

19

Out of 54 departments with material weekly sales.

DEPARTMENTS DECLINING 2010 → 2012

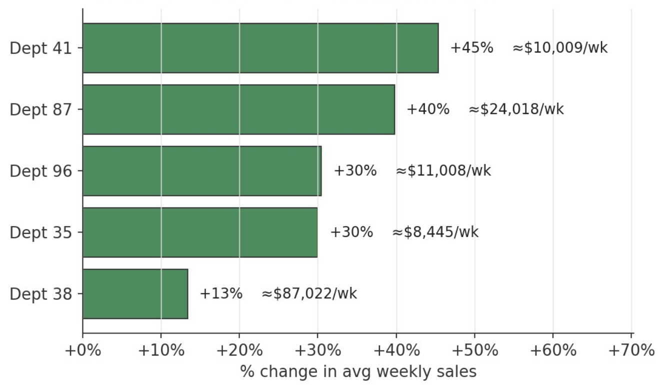
35

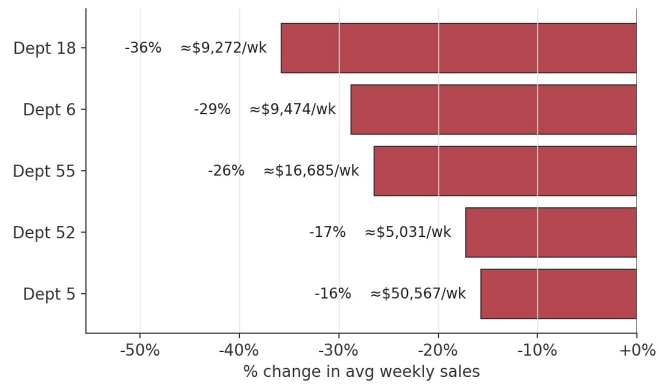
More departments are losing ground than gaining.

MEDIAN DEPARTMENT CHANGE

−2.0%

Consistent with the cohort-wide 2011 weakness.





Store 10 — top movers by average weekly sales, 2010 vs 2012 (Feb–Oct window, ≥ $5K/wk). Figures in parentheses are average weekly sales across the three years — they are the scale that gives the percentage its dollar weight.

THE DIAGNOSTIC READING OF THIS CHART

The two departments worth the most managerial attention are **Dept 5** (losing 16% off a $50,567/week base — the biggest absolute-dollar bleed in the store) and **Dept 38** (gaining 13% on an $87,022/week base — the biggest absolute-dollar gain). Those two alone move more money than the other eight combined. The other three decliners (Depts 18, 6, 55) are also worth investigating, particularly Dept 55, which loses $4.3K/week off a $16.7K base — a 26% fall suggests a category-specific or local competitive change, not routine noise.

07 Recommendations

Five actions, in decreasing order of urgency. The first two correct prior narrative; the last three are operational.

NARRATIVE · PRIORITY 1

Reframe the story of Store 10 for senior leadership

Replace the "high performer but declining" summary in Bob's original deck with "healthy, top-decile peer-adjusted performer with a flat top line and active internal reshuffling." The corrected framing is the one supported by the data; anything that carries forward the original framing will lead the client to wrong conclusions about Store 10's trajectory.

DASHBOARD · PRIORITY 1

Fix the year-over-year comparison logic company-wide

Any executive dashboard that compares year-to-date numbers against full prior-year numbers will produce the same phantom decline we found here, for many stores, not just Store 10. A "comparable-period mode" that restricts every year to the window available in the latest year is a straightforward, high-leverage fix.

OPERATIONAL · PRIORITY 2

Commission a focused review of Departments 5 and 55

These two account for the largest absolute-dollar losses in the store. Review should cover: (i) is the decline local to Store 10 or occurring across peer stores too? (ii) are there identifiable local competitive or demographic changes? (iii) have merchandising, pricing, or assortment decisions in these categories changed? The answer dictates whether this is a store issue, a category issue, or a market issue — each with a different response.

OPERATIONAL · PRIORITY 2

Codify what's working in Departments 38 and 87

Dept 38 is growing on a large base; Dept 87 is growing fastest. Understanding *why* (assortment changes, pricing, promotion cadence, local demand shift) produces a playbook that can be tested in other departments within Store 10 and, if it generalises, in peer stores too. Resilience in the 2011 downturn suggests the store has operational practices worth surfacing.

GOVERNANCE · PRIORITY 3

Adopt peer-adjusted KPIs for store performance reviews

The performance-vs-expectation residual used in this report is a more honest KPI than raw sales or raw sales growth. It controls for store size, type, local conditions, and seasonality, and it surfaces the genuine management signal. The model can be re-fit quarterly on the current window and the residual reported alongside existing metrics. Over time this sharpens every store-level conversation.

08 Methodology — a one-page note

This section exists so that nothing in the report needs to be taken on trust. The full pipeline, including code, is in the companion Jupyter notebook.

Data

Three provided datasets (store attributes, weekly sales by store–department, and weekly contextual features) joined and aggregated to the store–week level (6,435 rows). Missing values inspected and handled conservatively; no observations dropped.

Modelling

Two regression models were built to predict weekly store-level sales as a function of store size, type, temperature, fuel price, CPI, unemployment, holiday flag, month, and week of year.

- **Linear regression** — simple, interpretable baseline. Test MAE ≈ $225K, $R^2 \approx 0.72$.
- **Random Forest** — non-linear, handles seasonality and interactions. Test MAE ≈ $97K, $R^2 \approx 0.92$. Selected for the residual analysis.

Validation

A **time-based holdout** split trained models on pre-April 2012 data and evaluated on the subsequent holdout. This matches the intended use of the model (evaluate the most recent period against historical patterns) and avoids the leakage that random splits would introduce on temporally ordered data.

Peer cohort definition

Type B stores with size within ±25% of Store 10's 126,512 sq ft (10 stores). The ±25% band was chosen to give a statistically meaningful cohort ($n \geq 10$) while staying close to Store 10 on the two attributes most predictive of sales scale.

Honest caveats

The chosen model shows some overfitting (train $R^2 \approx 0.996$ vs test $R^2 \approx 0.92$); a production deployment would tighten tree depth and use rolling-window cross-validation. "Residual" measures performance-vs-peer, not good-vs-bad in an absolute sense. No causal claims are made — confirming that specific practices drive Store 10's outperformance would require a controlled test.

Source data: anonymized supermarket dataset provided with the case (Kaggle: Walmart Sales Forecast). Analysis performed in Python (pandas, scikit-learn, matplotlib). Reproducible pipeline in Bobs_Report.ipynb.